

#oops tasks

'''#1.online food delivery app

class Restaurant:

def __init__(self,menu):

self._menu=menu

def show_menu(self,*args):

print("Menu:",self._menu)

def prepare_food(self):

pass

class Pizza_restaurant(Restaurant):

def prepare_food(self):

print("preparing pizza")

class burger_restaurant(Restaurant):

def prepare_food(self):

print("preparing burger")

class south_indian_restaurant(Restaurant):

def prepare_food(self):

print("south indian food preparing")

choice = input("choose your restaurant(pizza/burger/south indian):")

if choice.lower() == "pizza":

obj_Pizza_restaurant(["veg pizza","non veg pizza","cheese pizza"])

elif choice.lower() == "burger":

obj_burger_restaurant(["veg burger","non veg burger","cheese burger"])

elif choice.lower() == "south indian":

```
obj_south_indian_restaurant(["dosa","idli","sambar","chutney"])
```

```
obj.show_menu()
```

```
obj.prepare_food()'''
```

```
'''#2.payment gateway system
```

```
from abc import ABC, abstractmethod
```

```
class payment_method(ABC):
```

```
    def __init__(self,account):
```

```
        self._account=account
```

```
    @abstractmethod
```

```
    def process_payment(self,amount):
```

```
        pass
```

```
class UPI(payment_method):
```

```
    def process_payment(self,amount):
```

```
        print(f"₹{amount} paid using UPI")
```

```
amount = float(input("enter your amount:"))
```

```
method =input("enter your method(UPI):")
```

```
if method == "UPI":
```

```
    obj = UPI("upi@bank")
```

```
obj.process_payment(amount)"""
```

```
"""#3.SMART HOME AUTOMATION
```

```
from abc import ABC, abstractmethod
```

```
class smartdevice(ABC):
```

```
    def __init__(self):
```

```
        self._status="OFF"
```

```
    def turn_on(self):
```

```
        self._status = "ON"
```

```
    def turn_off(self):
```

```
        self._status = "OFF"
```

```
    def get_status(self):
```

```
        return self._status
```

```
    @abstractmethod
```

```
    def operator(self):
```

```
        pass
```

```
class fan(smartdevice):
```

```
    def operator(self):
```

```
        print("fan is running")
```

```
def fridge(self):  
    print("fridge is operating")  
  
choice = input("choose your device(fridge/fan):")  
if choice.lower() == "fridge":  
    obj=fridge  
else :  
    obj=fan  
  
obj.turn_on()  
print("status:",obj.get_status())  
obj.operator()'''
```